

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_095743 | Base: main-v1.0

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Decision: GO_WITH_CONSTRAINTS | Confidence: 53%

CHANGE REQUEST

oscurare i pannelli

CASCADE ISSUES IDENTIFIED

[MAJOR] ISSUE-001 — Power margin reduction

Subsystem: power | Req: L1-PWR-001 | Type: constraint_violation

Removing the right solar panel reduces power generation capacity, potentially exceeding peak power consumption limits during imaging mode.

[MINOR] ISSUE-002 — Thermal stability impact

Subsystem: thermal | Req: L1-THM-001 | Type: missing_test

Reduced power generation may affect thermal control, increasing the risk of exceeding ΔT limits across orbital cycles.

[MINOR] ISSUE-003 — Structural balance alteration

Subsystem: mechanical | Req: None | Type: reanalysis_required

Removing the right solar panel alters the structural balance, potentially affecting mechanical stability and vibration response.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-31C §5.4 — Thermal balance test to validate TMM with reduced solar panel configuration.

Reason: To verify thermal stability after removing the right solar panel.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	53%	WARN	Thermal stability impact	■ MODIFIED
mechanical	80%	OK	Structural balance alteration	■ MODIFIED
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-PWR-001 — verification evidence invalidated, re-analysis required
- L1-THM-001 — verification evidence invalidated, re-analysis required